Questions

Q1.

A student rubs **two** balloons with a dry cloth.

The balloons become positively charged.

The student hangs the charged balloons using strings, as shown in Figure 5.

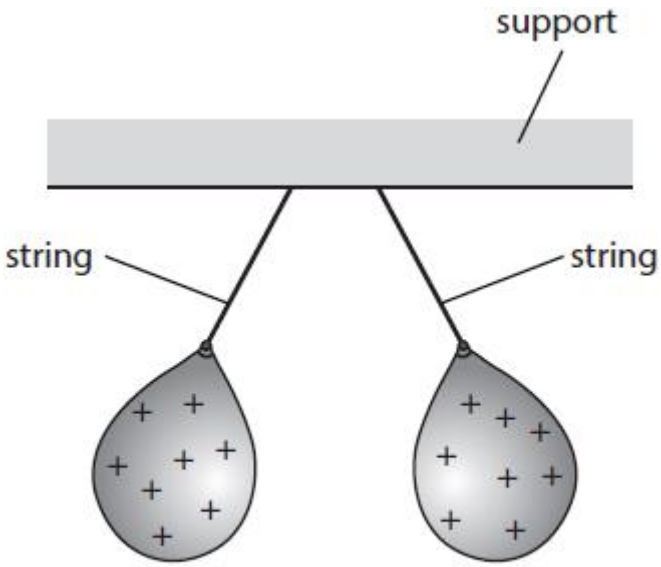

Figure 5

One of the charged balloons is moved so it nearly touches a wall, as shown in Figure 6.

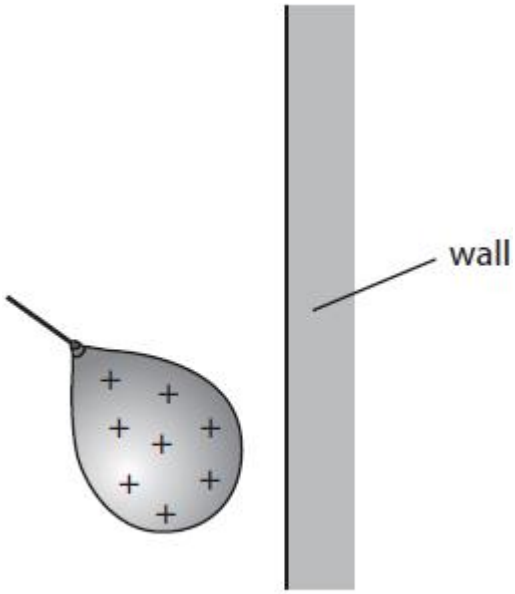

Figure 6

The balloon then sticks to the wall.

Explain why the balloon sticks to the wall.

You may add to the diagram to help your answer.

(2)

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(Total for question = 2 marks)

Q2.

* This question is about electrostatic charges and the forces between them.

Figure 16 shows some apparatus that can be used to show that like charges repel and unlike charges attract.

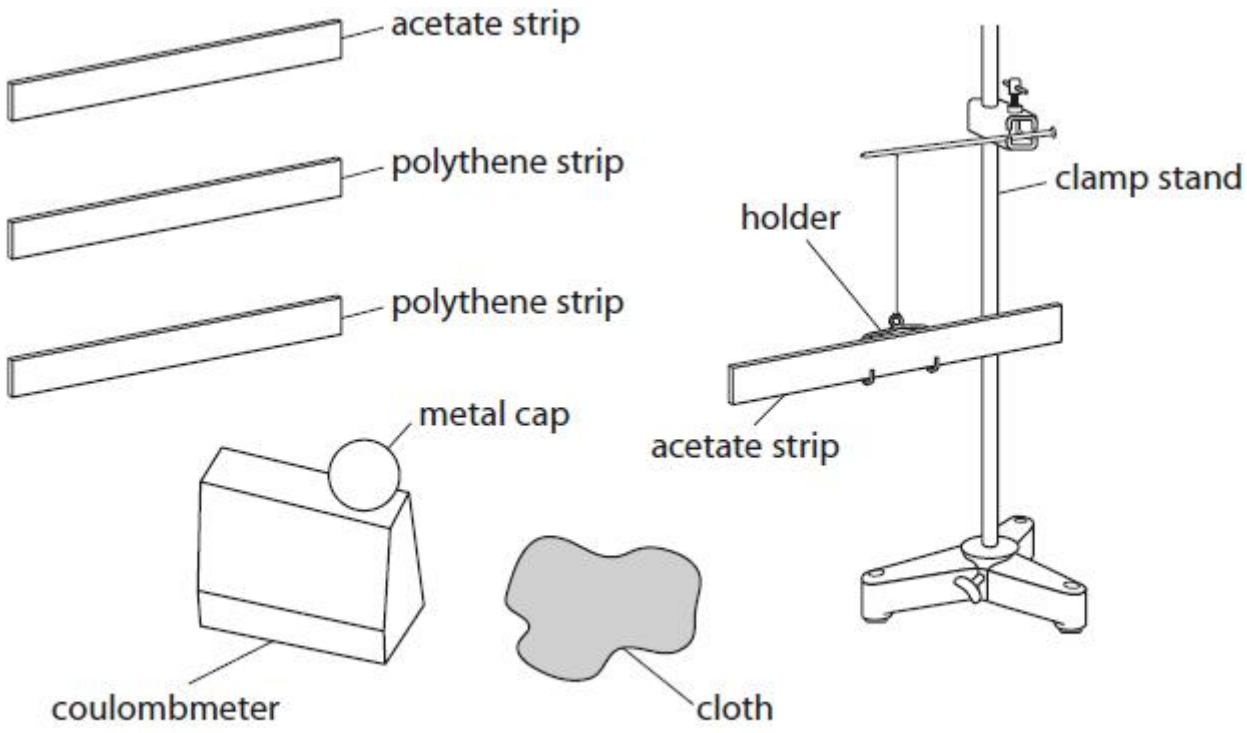

Figure 16

An acetate strip can be charged positively.

A polythene strip can be charged negatively.

A coulombmeter can be used to measure charge and whether the charge is positive or negative.

Explain how you would use the apparatus in Figure 16 to show that like charges repel and unlike charges attract.

(6)

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(Total for question = 6 marks)

Q3.

Figure 17 shows an airport worker refuelling an aircraft.

(Source: © Stanisław Tokarski/123RF)

Figure 17

As fuel moves through the pipe, it becomes positively charged.

Explain how the worker can prevent a build-up of charge when pumping fuel into the aircraft.

(3)

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(Total for question = 3 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A student uses a cloth to give a plastic rod a positive charge.

(i) Explain how the rod becomes positively charged.

(3)

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(ii) Figure 12 shows four light balls, Q, R, S and T.

Each ball is suspended on a nylon string.
Balls Q, R and T are coated with a conducting material.
Ball S is an insulator.
Q and S have no charge, R is positively charged and T is negatively charged.

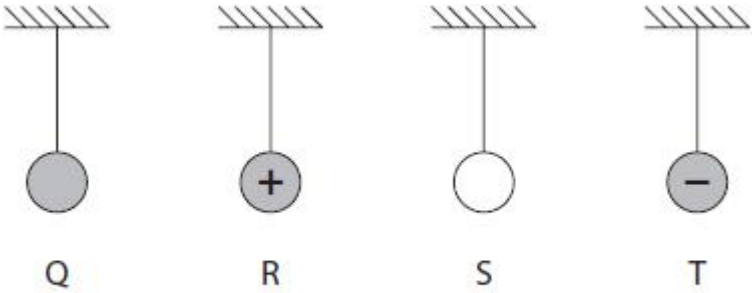

Figure 12

The student brings the positively charged rod near to each ball in turn.
Which ball is repelled by the positively charged rod?

(1)

- ☒ **A** Q
- ☒ **B** R
- ☒ **C** S
- ☒ **D** T

(Total for question = 4 marks)

Q5.

Figure 13 shows part of a cloud, above the ground.
The base of the cloud is negatively charged.

Figure 13

Explain how lightning is produced between the cloud and the Earth.
Your answer should refer to induced charges.
You may add to the diagram in Figure 13 to help your answer.

(3)

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(Total for question = 3 marks)

Q6.

This question is about static electricity.

Figure 2 shows a person touching a charged dome.

The person's hair is standing on end.

Figure 2

(i) Explain how electric charge causes the hair to spread out, as shown in Figure 2.

(2)

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(ii) The person lets go of the charged dome.

The charge on the person's hair is 10 μC.
The charge on the dome is 25 μC.
Calculate the percentage of charge on the hair compared with on the dome.
Use the equation

$$\text{percentage of charge on the hair} = \frac{\text{charge on hair}}{\text{charge on dome}} \times 100$$

(2)

percentage of charge on the hair = ... %

(Total for question = 4 marks)

Q7.

Explain how an aircraft can become electrically charged as it flies through the air.

(2)

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(Total for question = 2 marks)

Q8.

A student rubs **two** balloons with a dry cloth.

The balloons become positively charged.

The student hangs the charged balloons using strings, as shown in Figure 5.

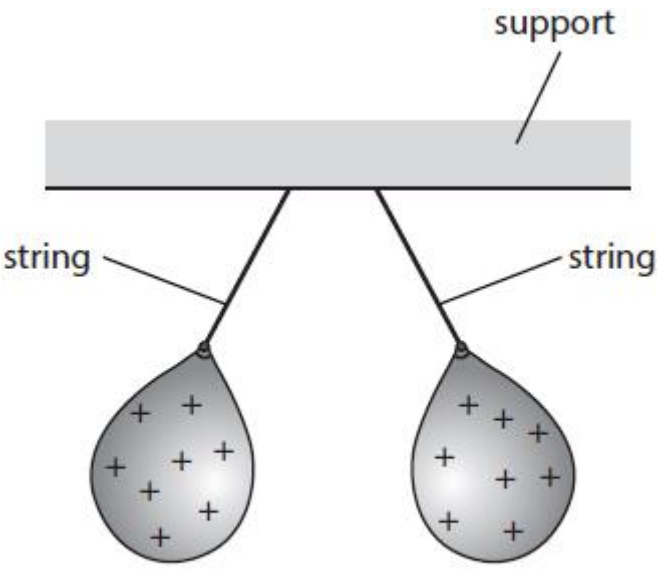

Figure 5

Figure 7 shows a positively charged metal sphere above the ground.

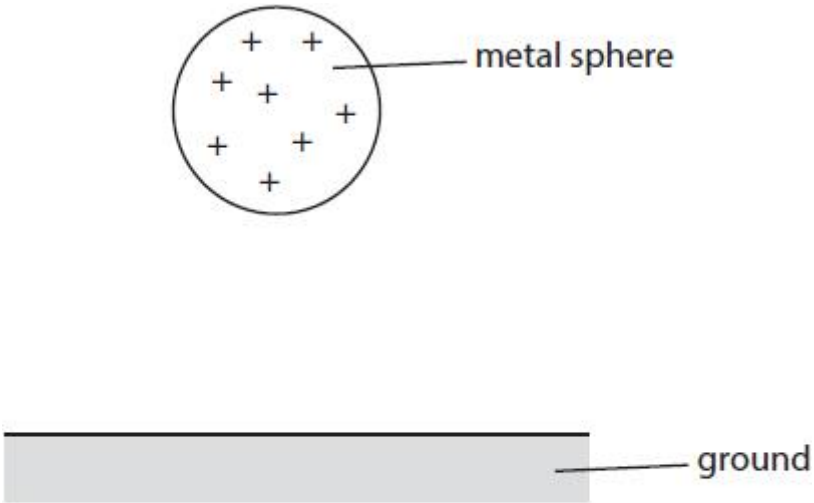

Figure 7

The metal sphere can be discharged by connecting the sphere to the ground with a metal wire.

Explain how this would discharge the sphere.

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(Total for question = 2 marks)

Q9.

An atom contains electrons, neutrons and protons.
*The photographs show some electrostatic effects.

positively charged balloon near hair

positively charged rod near some paper

positively charged balloon near a thin stream of water

Explain in terms of electric charges how one of these effects is caused.
You may include diagrams to help with your answers.

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Q10.

Figure 19 shows **two** metal plates.

The plates are charged by a very high voltage (potential difference).

Draw, on Figure 19, the shape and direction of the electric field between the two plates.

(2)

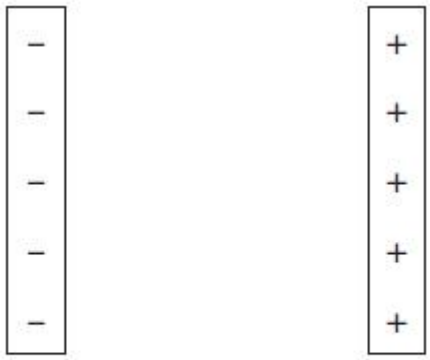


Figure 19

(Total for question = 2 marks)

Mark Scheme

Q1.

Question	Answer	Additional guidance	Mark
	an explanation linking any **two** from: balloon attracted (1) (by) negative charges/electrons (at surface) on wall (1) idea of charge separation in wall (1)	ignore 'sticks/attaches to the wall' as in stem may be drawn on Figure 6 eg minus signs induced charges (in wall) may be drawn on Figure 6	**(2)** **AO1.1**

Q2.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO2 (strand 2) (6 marks)** 1. rub a strip with the cloth 2. test to see its charge (positive or negative) by rubbing against the coulombmeter 3. place the strip in the holder 4. charge another strip by friction (rubbing) 5. test its charge with the meter 6. bring the second strip up towards the one in the holder; either attracts or repels 7. repeat 1-6 with another charged strip 8. repeat steps 1-3, but then charge an acetate strip, again detecting the sign of its charge, as before 9. bring the charged acetate (+) strip up to the charged polythene (-) strip; the one in the stirrup should now be attracted / move towards the other	**(6) AO2**

AO targeting: 6 marks AO2 strand 2

Level	Mark	Descriptor
	0	• No awardable content
Level 1	1-2	• The explanation attempts to link and apply knowledge and understanding of scientific enquiry, techniques and procedures, flawed or simplistic connections made between elements in the context of the question. • Lines of reasoning are unsupported or unclear. (AO2)
Level 2	3-4	• The explanation is mostly supported through linkage and application of knowledge and understanding of scientific enquiry, techniques and procedures, some logical connections made between elements in the context of the question. • Lines of reasoning mostly supported through the application of relevant evidence. (AO2)
Level 3	5-6	• The explanation is supported throughout by linkage and application of knowledge and understanding of scientific enquiry, techniques and procedures, logical connections made between elements in the context of the question. • Lines of reasoning are supported by sustained application of relevant evidence. (AO2)

Summary for guidance

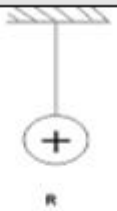
Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance A limited explanation with at least two steps of a suitable procedure	Possible candidate responses rub a strip with the cloth / give it a charge place the strip in the holder explains consequent attraction / repulsion
Level 2	3–4	Additional guidance Partially complete explanation of a suitable procedure with at least three steps and some purpose indicated	Possible candidate responses As above with bring a second strip up towards the one in the holder either attracts or repels repeated with like / unlike charges use of coulombmeter
Level 3	5–6	Additional guidance Detailed explanation of a suitable procedure with most steps and a clear logic in what is being proposed	Possible candidate responses As above with test charge with the meter to see if it's (+) or (-) detailed use of acetate / polythene strips with purpose

Q3.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification – knowledge (1 mark) and reasoning/justification – understanding (2 marks): • use of a conductor to connect between aircraft and ground (1) • allowing negative charge to move onto the aircraft (1) • therefore neutralising the positive charge(s) (1)	accept (copper) wire accept earth for ground	(3)

Q4.

Question Number	Answer	Additional guidance	Mark
(i)	An explanation that combines:-		(3)
	rub the rod with a cloth (1)	allow clean off the rod or friction (with the rod)	
	(so)electrons (1)	allow <u>negative</u> charges for electrons	
	are moved (from rod to cloth) (1)	movement of <u>positive</u> charges can only score the first mark 'electrons are positive' can score a maximum of one mark movement of unnamed charges can score third mark	

Question Number	Answer	Mark
(ii)	**B** R  **B** is the only correct answer. **A** is incorrect because ball Q is coated with a conducting material but is uncharged, a negative charge will be induced on it and it will be attracted not repelled by a positively charged rod. **C** is incorrect because ball S is an insulator and is uncharged and will not be repelled by a positively charged rod. **D** is incorrect because ball T has a negative charge and will be attracted not repelled by a positively charged rod.	**(1)**

Q5.

Question Number	Answer	Additional guidance	Mark
	An explanation that includes any three of the following points :-	May be seen on diagram	**(3)**
	ground is charged (by induction) (1)		
	charge on ground is positive (1)	Award two marks for 'the ground is positively charged'	
	electric field builds up (between cloud and ground) (1)	allow electric charge or voltage or potential difference for electric field	
	air is ionised (1)	air becomes a conductor allow charge for ions	
	electrons travel to the ground/positive ions travel to the cloud (1)		

Q6.

Question number	Answer	Additional guidance	Mark
(i)	explanation linking like/same charges (on strands of hair) (1) (like charges) repel (1)	 positives / protons negatives / electrons if no other mark allow one mark for charge / 'it' / electron(s) moves OR current (in body, to or from dome) ignore 'static'	(2) AO1

Question number	Answer	Additional guidance	Mark
(ii)	substitution (1) (%) = $\frac{10}{25}$ (x 100) evaluation (1) (%) = 40 (%)	 accept 0.4 accept 10 and 25 written next to numerator and denominator of the stated equation award full marks for the correct answer without working	(2) AO2

Q7.

Question number	Answer	Additional guidance	Mark
	An explanation that combines identification – understanding (1 mark) and reasoning/justification – understanding (2 marks): • there is friction between aircraft and air (1) • which causes electron transfer between aircraft and air (1)	accept idea of air rubbing against wings ignore "charge" "static" do not allow (for second mark) idea of protons moving	(2)

Q8.

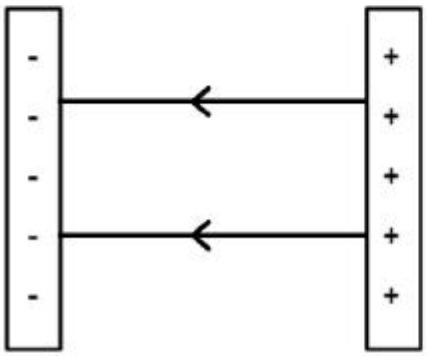
Question	Answer	Additional guidance	Mark
	an explanation linking **two** from:	explanations using moving positive charges OR moving positive electrons can score a maximum of ONE mark for this item	**(2) AO1.1**
	metal/wire is a conductor (1)		
	charge (electrons) moves in the wire (1)	if no other marks scored allow protons or positive charge(s) (in this context) moves <u>to the ground</u> for this mark only	
	electrons/negative charges (move) from ground to the sphere (1)		
	(the sphere is) earth(ed) (1)	(the sphere is) ground(ed) OR electrons neutralise (the positive charge on) the sphere	

Q9.

		Indicative Content	Mark
		An explanation etc. including some of the following points • static electricity • opposites charges attract • charges are different • induced charges • charges separate • charges move • electrons move	**(6)**

		• electrons move towards a positive charge / balloon / rod Allow credit for a correct explanation for an effect which is not given in the question. Allow credit for separation of charge being shown on a diagram.	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited explanation. Explains the effect is caused by charges. e.g. the charge on the balloon pulls the water; the charge on the rod attracts the bits of paper; the balloon is rubbed to give it charge; opposites attract; positive and negative attract; • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple explanation. Explains an effect is caused by opposite charges attracting or like charges repelling. e.g. the charge on the balloon is opposite to the charge on the water so they attract; the positive charges on the balloon attract negative charges on the girl's hair; • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed explanation. Explains the effect is caused by induction, charge separation or moving electrons which leads to attraction between opposite charges. e.g. the electrons have been moved off the balloon so it has a positive charge and attracts the negative charge on the hair; the balloon has a positive charge and induces a negative charge on the stream of water which attracts it; • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q10.

Question number	Answer	Additional guidance	Mark
	at least two horizontal straight lines between the two plates, judge by eye (1) arrows **only** from +ve towards -ve (1)	independent marks lines do not have to touch plates ignore curved lines at top and bottom of plate(s) example, scores 2 marks 	**(2)** **AO1**